

# Scenario 2 output 2 ollama

Here is the analysis of potential threats to the LLM-integrated secure file management system:

## **\*\*STRIDE Threats\*\***

### 1. **\*\*Threat ID:\*\* T001**

**\*\*Threat name:\*\* Unauthorized File Access**

**\*\*Category:\*\* Data leakage**

**\*\*Affected component/data flow:\*\* DS1, P2**

**\*\*Attack description:\*\* An attacker can exploit a vulnerability in P2 to bypass authorization checks and access unauthorized files.**

**\*\*Impact:\*\* High – sensitive information may be exposed.**

**\*\*Severity:\*\* 9/10 (high impact, moderate severity)**

**\*\*Likelihood:\*\* Medium – dependent on implementation and security controls.**

**\*\*Mitigation:\*\* Implement robust security controls, such as encryption and secure coding practices.**

### 2. **\*\*Threat ID:\*\* T002**

**\*\*Threat name:\*\* Broken Access Control**

**\*\*Category:\*\* Authorization threats**

**\*\*Affected component/data flow:\*\* P2, DS1**

**\*\*Attack description:\*\* An attacker can exploit a vulnerability in P2 to manipulate access control decisions.**

**\*\*Impact:\*\* High – unauthorized users may access sensitive information.**

**\*\*Severity:\*\* 9/10 (high impact, moderate severity)**

**\*\*Likelihood:\*\* Medium – dependent on implementation and security controls.**

**\*\*Mitigation:\*\* Implement robust access control mechanisms, such as role-based access control.**

### 3. **\*\*Threat ID:\*\* T003**

**\*\*Threat name:\*\* Data Tampering**

**\*\*Category:\*\* Data leakage**

**\*\*Affected component/data flow:\*\* DS1**

**\*\*Attack description:\*\* An attacker can exploit a vulnerability in the system to modify or delete sensitive information.**

**\*\*Impact:\*\* High – data integrity and confidentiality may be compromised.**

**\*\*Severity:\*\* 9/10 (high impact, moderate severity)**

**\*\*Likelihood:\*\* Medium – dependent on implementation and security controls.**

**\*\*Mitigation:\*\* Implement robust security controls, such as encryption and secure coding practices.**

### 4. **\*\*Threat ID:\*\* T004**

**\*\*Threat name:\*\* Elevation of Privilege**

**\*\*Category:\*\* Authorization threats**

**\*\*Affected component/data flow:\*\* P2, EE1**

**\*\*Attack description:\*\* An attacker can exploit a vulnerability in the system to elevate their privileges.**

**\*\*Impact:\*\*** High – unauthorized users may access sensitive information or modify critical systems.  
**\*\*Severity:\*\*** 9/10 (high impact, moderate severity)  
**\*\*Likelihood:\*\*** Low – dependent on implementation and security controls.  
**\*\*Mitigation:\*\*** Implement robust security controls, such as secure coding practices and least privilege principle.

5. **\*\*Threat ID:\*\*** T005

**\*\*Threat name:\*\*** Session Hijacking

**\*\*Category:\*\*** Data leakage

**\*\*Affected component/data flow:\*\*** EE1, DS1

**\*\*Attack description:\*\*** An attacker can exploit a vulnerability in the system to hijack user sessions.

**\*\*Impact:\*\*** Medium – sensitive information may be exposed or modified.

**\*\*Severity:\*\*** 7/10 (medium impact, low severity)

**\*\*Likelihood:\*\*** Low – dependent on implementation and security controls.

**\*\*Mitigation:\*\*** Implement robust security controls, such as secure session management.

**\*\*LLM-specific threats\*\***

1. **\*\*Threat ID:\*\*** T006

**\*\*Threat name:\*\*** LLM Context Manipulation

**\*\*Category:\*\*** Data leakage

**\*\*Affected component/data flow:\*\*** P3, EE2

**\*\*Attack description:\*\*** An attacker can exploit a vulnerability in the system to manipulate context sent to the external LLM provider.

**\*\*Impact:\*\*** High – sensitive information may be exposed or modified.

**\*\*Severity:\*\*** 9/10 (high impact, moderate severity)

**\*\*Likelihood:\*\*** Low – dependent on implementation and security controls.

**\*\*Mitigation:\*\*** Implement robust security controls, such as secure coding practices and input validation.

2. **\*\*Threat ID:\*\*** T007

**\*\*Threat name:\*\*** LLM Output Manipulation

**\*\*Category:\*\*** Data leakage

**\*\*Affected component/data flow:\*\*** P3, EE2

**\*\*Attack description:\*\*** An attacker can exploit a vulnerability in the system to manipulate output received from the external LLM provider.

**\*\*Impact:\*\*** Medium – sensitive information may be exposed or modified.

**\*\*Severity:\*\*** 7/10 (medium impact, low severity)

**\*\*Likelihood:\*\*** Low – dependent on implementation and security controls.

**\*\*Mitigation:\*\*** Implement robust security controls, such as secure coding practices and input validation.

3. **\*\*Threat ID:\*\*** T008

**\*\*Threat name:\*\*** LLM Provider Risks

**\*\*Category:\*\*** External Entity risks

**\*\*Affected component/data flow:\*\*** EE2

**\*\*Attack description:\*\*** An attacker can exploit a vulnerability in the external LLM provider to access sensitive information.

**\*\*Impact:\*\*** High – sensitive information may be exposed or modified.  
**\*\*Severity:\*\*** 9/10 (high impact, moderate severity)  
**\*\*Likelihood:\*\*** Low – dependent on implementation and security controls.  
**\*\*Mitigation:\*\*** Implement robust security controls, such as secure coding practices and input validation.

**\*\*OWASP LLM Top 10 threats\*\***

1. **\*\*Threat ID:\*\*** T009

**\*\*Threat name:\*\*** Insecure Data Context Handling

**\*\*Category:\*\*** Insecure context handling

**\*\*Affected component/data flow:\*\*** P3, EE2

**\*\*Attack description:\*\*** An attacker can exploit a vulnerability in the system to manipulate sensitive information sent to the external LLM provider.

**\*\*Impact:\*\*** High – sensitive information may be exposed or modified.

**\*\*Severity:\*\*** 9/10 (high impact, moderate severity)

**\*\*Likelihood:\*\*** Low – dependent on implementation and security controls.

**\*\*Mitigation:\*\*** Implement robust security controls, such as secure coding practices and input validation.

2. **\*\*Threat ID:\*\*** T010

**\*\*Threat name:\*\*** Sensitive Document Exposure

**\*\*Category:\*\*** Data leakage

**\*\*Affected component/data flow:\*\*** DS1

**\*\*Attack description:\*\*** An attacker can exploit a vulnerability in the system to access sensitive documents.

**\*\*Impact:\*\*** High – sensitive information may be exposed or modified.

**\*\*Severity:\*\*** 9/10 (high impact, moderate severity)

**\*\*Likelihood:\*\*** Low – dependent on implementation and security controls.

**\*\*Mitigation:\*\*** Implement robust security controls, such as encryption and secure coding practices.

3. **\*\*Threat ID:\*\*** T011

**\*\*Threat name:\*\*** Insecure LLM Context Handling

**\*\*Category:\*\*** Insecure context handling

**\*\*Affected component/data flow:\*\*** P3, EE2

**\*\*Attack description:\*\*** An attacker can exploit a vulnerability in the system to manipulate sensitive information sent to the external LLM provider.

**\*\*Impact:\*\*** High – sensitive information may be exposed or modified.

**\*\*Severity:\*\*** 9/10 (high impact, moderate severity)

**\*\*Likelihood:\*\*** Low – dependent on implementation and security controls.

**\*\*Mitigation:\*\*** Implement robust security controls, such as secure coding practices and input validation.

**\*\*File security threats\*\***

1. **\*\*Threat ID:\*\*** T012

**\*\*Threat name:\*\*** Malicious Document Content

**\*\*Category:\*\*** Data leakage

**\*\*Affected component/data flow:\*\*** DS1

**\*\*Attack description:\*\*** An attacker can exploit a vulnerability in the system to upload malicious documents.  
**\*\*Impact:\*\*** High – sensitive information may be exposed or modified.  
**\*\*Severity:\*\*** 9/10 (high impact, moderate severity)  
**\*\*Likelihood:\*\*** Low – dependent on implementation and security controls.  
**\*\*Mitigation:\*\*** Implement robust security controls, such as encryption and secure coding practices.

## 2. **\*\*Threat ID:\*\*** T013

**\*\*Threat name:\*\*** Prompt Injection through Documents

**\*\*Category:\*\*** Data leakage

**\*\*Affected component/data flow:\*\*** DS1, P3

**\*\*Attack description:\*\*** An attacker can exploit a vulnerability in the system to inject malicious prompts through documents.

**\*\*Impact:\*\*** High – sensitive information may be exposed or modified.

**\*\*Severity:\*\*** 9/10 (high impact, moderate severity)

**\*\*Likelihood:\*\*** Low – dependent on implementation and security controls.

**\*\*Mitigation:\*\*** Implement robust security controls, such as secure coding practices and input validation.

## 3. **\*\*Threat ID:\*\*** T014

**\*\*Threat name:\*\*** Unauthorized File Access

**\*\*Category:\*\*** Data leakage

**\*\*Affected component/data flow:\*\*** DS1, P2

**\*\*Attack description:\*\*** An attacker can exploit a vulnerability in the system to access unauthorized files.

**\*\*Impact:\*\*** High – sensitive information may be exposed or modified.

**\*\*Severity:\*\*** 9/10 (high impact, moderate severity)

**\*\*Likelihood:\*\*** Medium – dependent on implementation and security controls.

**\*\*Mitigation:\*\*** Implement robust security controls, such as encryption and secure coding practices.

**Note:** The severity and likelihood of the threats are subjective and may vary depending on the specific implementation and context.